CBS COPENHAGEN BUSINESS SCHOOL

FAROE ISLANDS 2030: NO CABLE, ICELAND OR SCOTLAND

Extra yearly cost of a 100 % renewable main grid at 2030 demand, without a cable and with an import-only cable of fixed or model-chosen size

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2030 demand is 19 % higher, and the cable only imports

Group assumptions behind every run

What we assume

- **One grid, only renewables:** main grid as one node, no oil in any hour; Suðuroy (about 7 % of demand) not modelled
- **Cable: import only,** no export from the Faroes (group decision 28.10.)
- **One data year:** demand (SEV) and weather (Copernicus C3S), Oct 2025-Sep 2026
- **2030 demand** = 600/506 x today's hourly profile (factor 1.186)
- **Brownfield:** today's plants stay; the model adds wind, solar, storage, cable

Extra cost per year = cost on top of today's system: annuities of new capacity + cable + the bill for imported power.

Demand in 2030

578 GWh

electricity demand per year (today 488 GWh)

93 MW

peak demand in the highest hour (today 78 MW)

Hydro inflow, biogas, wind and solar profiles and the cable prices are the same as today.

Sources: SEV, Copernicus C3S, Elma, Elexon, DEA; demand factor 600/506 = group assumption

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Nine cases: no cable, two cables, fixed or chosen size

Scenario grid; every run at 2030 demand and again at today's demand, cable for import only

	Length and price on the other side	Fixed size	Size chosen by the model
No cable (base)	No connection: 100 % renewable and self-sufficient	0 MW	-
Iceland	450 km; hourly Elma price, mean 27 EUR/MWh	25, 50 and 100 MW	0 to 1,000 MW
Scotland	400 km (NAEN 2016); GB day-ahead price, mean 111 EUR/MWh	25, 50 and 100 MW	0 to 500 MW (1,000 MW gives the same)

Cable cost (DEA): 3,220 EUR per MW and km + 640 k EUR per MW for converters, 40 years, 5 %, incl. O&M = 168 k EUR (Iceland) and 154 k EUR (Scotland) per MW per year. Model-chosen: cable MW is a decision like wind MW.

Sources: cable lengths NAEN (2016), North Atlantic Energy Network, p. 9; cable cost DEA technology catalogue; prices Elma, Elexon

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Iceland is cheaper than Scotland at every cable size

All runs at 2030 demand, import only: cable size, extra cost per year, new capacity and import

Case	Cable (MW)	Extra cost (M EUR/yr)	New wind (MW)	New solar (MW)	New battery (MW / MWh)	Import (GWh)	Import (% of demand)	Curtailed (%)*
No cable (base)	0	**24.8**	90	72	46 / 1,052	0	0 %	44 %
Iceland, fixed	25	**12.4**	29	8	15 / 362	45	8 %	15 %
Iceland, fixed	50	**11.1**	0	0	0	118	20 %	1 %
Iceland, fixed	100	**19.4**	0	0	0	118	20 %	1 %
Iceland, model-chosen	35	**9.3**	0	0	5 / 76	113	20 %	0 %
Scotland, fixed	25	**15.6**	39	0	21 / 358	44	8 %	22 %
Scotland, fixed	50	**17.3**	23	0	5 / 70	76	13 %	15 %
Scotland, fixed	100	**24.7**	20	0	12 / 103	80	14 %	14 %
Scotland, model-chosen	31	**15.1**	31	0	11 / 154	61	10 %	19 %

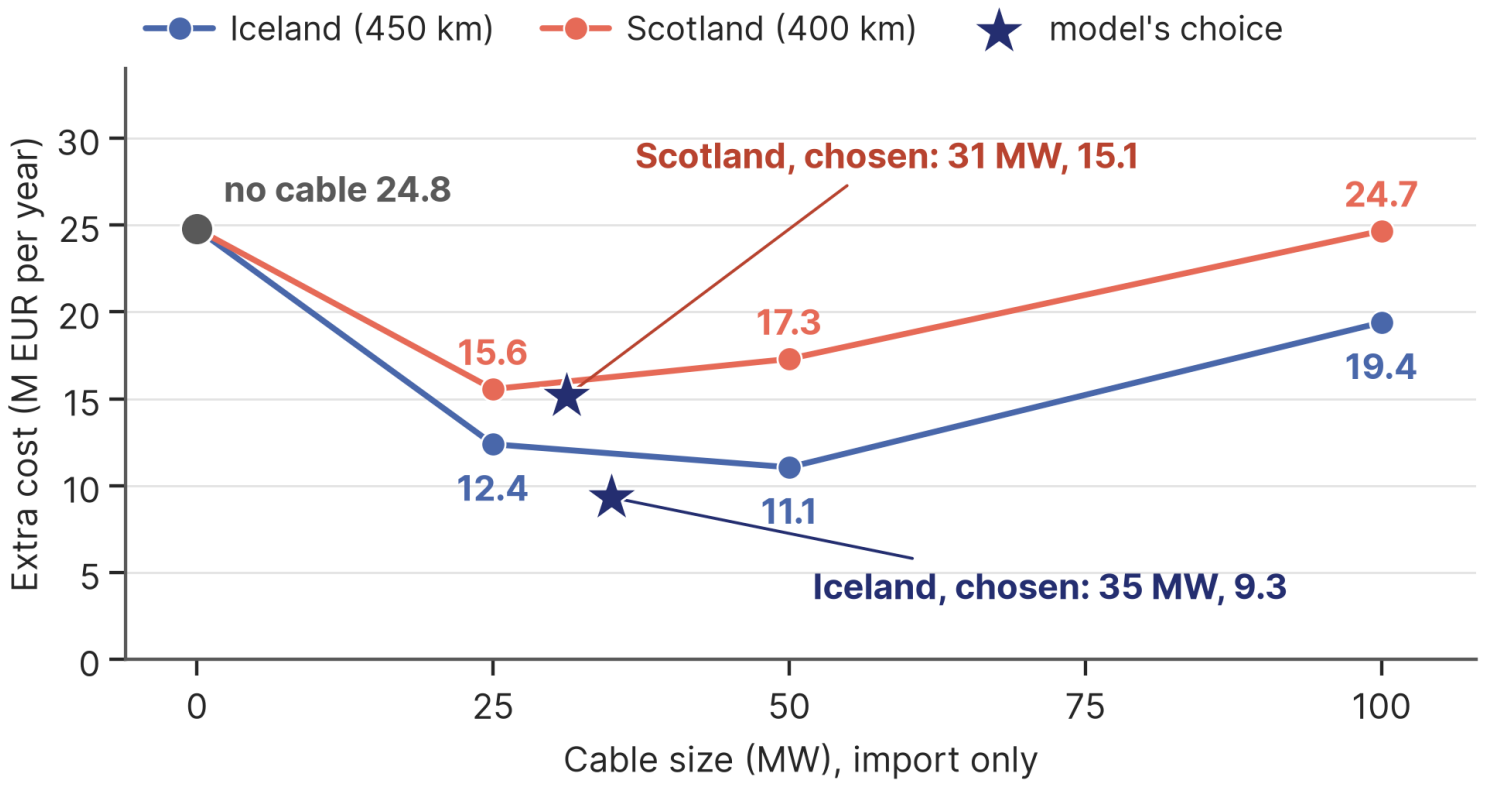
* share of available wind and solar power that is not used. No export in any run. Model-chosen sizes stay far below our upper limits (1,000 MW Iceland, 500 MW Scotland). No new pumped hydro in any run.

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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Both cables are cheapest at about 30 to 35 MW

Extra cost per year over cable size, 2030 demand, import only: fixed sizes and the model's choice



What it shows

- No cable: **24.8 M EUR** per year
- Iceland: lowest at 35 MW (**9.3**); at 100 MW 19.4
- Scotland: lowest at 31 MW (**15.1**); at 100 MW 24.7, about the same as no cable

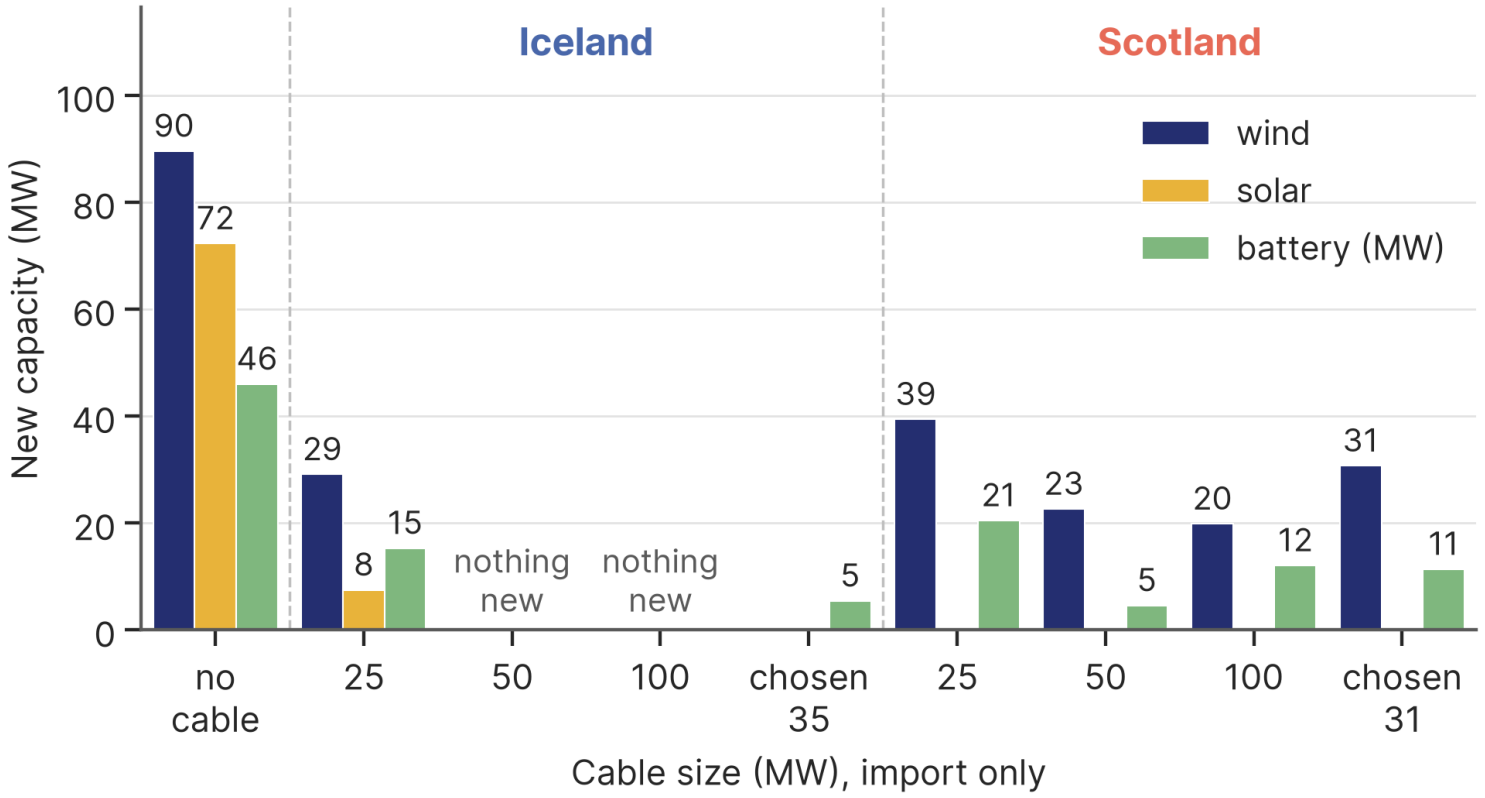
Without export income, every MW above the need only adds cable cost

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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A cable replaces most new wind, solar and storage

New capacity per run (MW), 2030 demand, import only



What it shows

- No cable: 90 MW wind, 72 MW solar, battery 46 MW / 1,052 MWh
- Iceland from 50 MW: nothing new; import 118 GWh (20 % of demand)
- Scotland: still 20 to 39 MW new wind, as GB power is dear

Average import price: Iceland about 22, Scotland about 100 EUR/MWh

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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Iceland: the model picks 35 MW at 9.3 M EUR per year

Fixed cable sizes compared with the size chosen by the model, Iceland, 2030 demand, import only

35 MW
cable size chosen by the model

9.3 M EUR
extra cost per year (no cable: 24.8)

1.7 M EUR
less than the best fixed size (50 MW, 11.1)

Cable size	Cable (MW)	Extra cost (M EUR/yr)	New wind (MW)	New battery (MW / MWh)	Import (GWh)	Import price (EUR/MWh)	Average cable use
fixed	25	**12.4**	29	15 / 362	45	26	20 %
fixed	50	**11.1**	0	0	118	23	27 %
fixed	100	**19.4**	0	0	118	22	13 %
model-chosen	35	**9.3**	0	5 / 76	113	22	37 %

- From 50 MW on the cable covers the whole gap: no new plants, 118 GWh import
- 100 MW imports the same as 50 MW; the extra 50 MW only add 8.3 M EUR cable cost per year

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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Scotland: the model picks 31 MW at 15.1 M EUR per year

Fixed cable sizes compared with the size chosen by the model, Scotland, 2030 demand, import only

31 MW
chosen size, far below our 500 MW limit

15.1 M EUR
extra cost per year (Iceland chosen: 9.3)

100 EUR
average price per MWh imported (Iceland: 22)

Cable size	Cable (MW)	Extra cost (M EUR/yr)	New wind (MW)	New battery (MW / MWh)	Import (GWh)	Import price (EUR/MWh)	Average cable use
fixed	25	**15.6**	39	21 / 358	44	100	20 %
fixed	50	**17.3**	23	5 / 70	76	92	17 %
fixed	100	**24.7**	20	12 / 103	80	79	9 %
model-chosen	31	**15.1**	31	11 / 154	61	100	22 %

- GB power is about four times as dear as Icelandic power, so the model still builds new wind
- Every Scotland case costs more than the same Iceland case (3.2 to 6.2 M EUR per year)

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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2030 demand makes every case more expensive than today

Extra cost per year (M EUR), import only, at today's demand (488 GWh) and at 2030 demand (578 GWh)

Case	Today	2030	Change
No cable (base)	16.9	**24.8**	+7.9
Iceland 25 MW	6.3	**12.4**	+6.1
Iceland 50 MW	9.9	**11.1**	+1.2
Iceland 100 MW	18.2	**19.4**	+1.2
Iceland, chosen (25 / 35 MW)	6.2	**9.3**	+3.1
Scotland 25 MW	10.4	**15.6**	+5.2
Scotland 50 MW	13.6	**17.3**	+3.7
Scotland 100 MW	21.2	**24.7**	+3.5
Scotland, chosen (22 / 31 MW)	10.1	**15.1**	+5.0

What it shows

- No cable: +7.9 M EUR per year for 91 GWh more demand
- Chosen cables grow: Iceland 25 to 35 MW, Scotland 22 to 31 MW
- Today a 100 MW cable is dearer than no cable (Iceland 18.2, Scotland 21.2 vs 16.9)

Cheapest case at both demand levels: Iceland cable of model-chosen size

Source: own model, Oct 2025-Sep 2026 inputs (SEV, Copernicus C3S, Elma, Elexon, DEA)

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The numbers leave out several real-world limits

Simplifications in the model and what they do to the results

1. **Linear cable cost per MW**
   No fixed cost per cable, so small cables look cheaper than in reality

2. **Cable always available**
   No faults or repair time; subsea repairs can take months

3. **Any amount at the hourly price**
   Iceland or Scotland deliver any amount up to the cable size

4. **GB power is mostly gas**
   Import from Scotland is often fossil power at the margin

5. **Flat demand scaling**
   Growth from heating and transport is in winter; we add it to summer too

6. **Suðuroy not modelled**
   Separate grid with about 7 % of demand, outside our numbers

7. **No export**
   Surplus wind is curtailed, never sold; 44 % of it in the no-cable case

Points 1 to 3 favour the cable cases; point 5 likely understates winter demand in all cases.